How Should AI Assist Ambiguous Data Decisions? Designing LLM Support for Table Unionability

NINA KLIMENKOVA, Worcester Polytechnic Institute, USA

Table unionability is often treated as a technical retrieval or matching problem, but in practice it also requires human judgment, especially when tables appear structurally similar yet differ in semantic meaning or context. At the same time, large language models (LLMs) are increasingly capable of providing semantic recommendations in data tasks, but it remains unclear how such assistance should be designed to support human reasoning rather than encourage overreliance. In this paper, we present the design of an LLM-assisted interface for table unionability assessment and outline an evaluation of how different forms of AI support shape human decision-making in ambiguous cases. Our system follows a two-stage workflow in which users first make an independent unionability judgment and then view one of several forms of LLM assistance, including a recommendation alone, a recommendation with explanation, or a recommendation with explanation and uncertainty cues. We propose to examine whether explanation format influences the quality of users' justifications and whether human-AI disagreement can serve as a signal of ambiguity in semantic data integration tasks. We also examine how these interface designs affect accuracy, confidence calibration and answer revision between humans and AI.This work contributes a human-centered design perspective on table unionability and aims to inform the development of interactive AI systems that support more reflective and appropriately calibrated decision-making in data discovery workflows.

CCS Concepts: • **Human-centered computing → Human computer interaction (HCI)**; *Interactive systems and tools*; *Empirical studies in HCI*; • **Information systems → Data Discovery**; *Table Unionability*.

Additional Key Words and Phrases: Human-AI Interaction, Large Language Models, Explainable AI, appropriate reliance, decision support, table unionability, data integration, ambiguity, uncertainty, semantic data discovery

ACM Reference Format:
Nina Klimenkova. 2018. How Should AI Assist Ambiguous Data Decisions? Designing LLM Support for Table Unionability. In *Proceedings of Make sure to enter the correct conference title from your rights confirmation email (Conference acronym 'XX).* ACM, New York, NY, USA, 9 pages. https://doi.org/XXXXXXX.XXXXXXX

1 Introduction

Table unionability is commonly framed as a technical problem in data integration and table search [4, 9, 10]. In practice, however, it is also a human judgment task [12]. Two tables may appear compatible at the schema level while differing in meaning, context, scope, or granularity, which makes many unionability decisions inherently ambiguous. This ambiguity is especially important in realistic data discovery settings, where deciding whether two tables should be combined is not always reducible to purely syntactic or statistical similarity. Prior work has largely focused on improving automated methods for unionability detection [4, 9, 10], while recent evidence suggests that both humans and LLMs struggle on difficult or borderline cases [12]. This makes unionability a useful setting for studying how AI can support human reasoning without replacing it.

Author's Contact Information: Nina Klimenkova, nklimenkova@wpi.edu, Worcester Polytechnic Institute, Worcester, MA, USA.

In this paper, we study table unionability as a human–AI interaction problem. Rather than treating the LLM as a substitute decision-maker, we design a pilot interface that preserves independent human judgment by requiring users to first provide an initial decision before seeing AI assistance. The interface then introduces one of several assistance conditions, including a (1) recommendation alone, a (2) recommendation with an explanation, or a (3) recommendation with explanation and uncertainty cues. This design enables us to examine not only whether LLM assistance improves outcomes, but also how the form of assistance shapes decision quality, confidence calibration, and users' willingness to revise their initial judgments. More broadly, we use this setting to study *appropriate reliance*: whether users engage with AI support in a way that helps them improve decisions without simply deferring to the model.

The paper addresses the following research questions:

- **RQ1.** What interface design enables the study of LLM assistance in ambiguous unionability tasks without replacing independent human judgment?
- **RQ2.** Do explanations and uncertainty cues improve decision quality and confidence calibration compared to recommendation-only assistance?
- **RQ3.** How does LLM assistance influence users' willingness to revise their initial judgments?

To further inform the study design, we conducted a pilot deployment of the interface and collected qualitative feedback from participants about the clarity, usefulness, and perceived influence of different forms of LLM assistance. We also carried out a preliminary analysis of the pilot data, which suggested that explanations provided by LLM may affect users' confidence and answer revision behavior differently than recommendation-only assistance. While these pilot observations are not intended as final evidence, they will help to refine the interface and provided initial indications that the form of AI assistance may meaningfully shape user behavior. We describe these pilot findings in Section 4. Together, the research questions and these early observations motivate the following hypotheses:

- **H1.** Compared to participants' initial unassisted judgments, LLM assistance will improve decision quality on unionability tasks.
- **H2.** Compared to recommendation-only assistance, assistance that includes explanations and uncertainty cues will lead to better decision quality and better confidence calibration.
- **H3.** LLM assistance will influence users' willingness to revise their initial judgments, with richer forms of assistance encouraging more selective and appropriate revision behavior.

By addressing these questions and hypotheses, this work contributes a human-centered perspective on table unionability and offers design insight into how interactive AI systems can support semantic data discovery in ambiguous decision settings.

2 Related Work

This project connects three lines of research: human judgment in matching data integration and discovery, appropriate reliance in human–AI decision-making, and prior work on table unionability.

Research in data integration has shown that matching is not purely a technical process, but also a human decision task shaped by cognitive and behavioral factors such as biases, consistency, and confidence [1, 15]. Ackerman et al. [1] showed that human matching decisions can exhibit predictable biases and inconsistency, suggesting that ambiguity in data integration tasks should be studied not only as an algorithmic challenge but also as a judgment problem. This perspective is important for unionability, where users may need to interpret semantic similarity, contextual compatibility, and granularity differences rather than apply a fixed rule.

A second line of work examines *appropriate reliance* in human–AI decision-making. Prior studies show that people do not simply follow AI advice, but reconcile it with their own beliefs, intuitions, and task understanding. Chen et al. [7] found that users' reliance on AI explanations depends on how those explanations interact with their own intuition, and that some explanation forms can even increase overreliance. Similarly, Zhang et al. [16] showed that confidence information can improve trust calibration, while explanations alone do not necessarily improve decision quality. More recently, Cao et al. [6] showed that appropriate reliance depends not only on whether uncertainty is shown, but also on *how* it is presented and how it interacts with users' initial judgments. Together, these studies suggest that the effectiveness of AI assistance depends on interface design, not only on model output.

Related work has also explored interventions that help users avoid blindly accepting AI advice. Buçınca et al. [5] showed that cognitive forcing functions can reduce overreliance on AI recommendations, while Ma et al. [11] showed that appropriate trust improves when users are supported in reasoning about both AI correctness and their own likely correctness. At a broader level, Bansal et al. [3] argued that the most accurate AI is not necessarily the best teammate, emphasizing that human–AI performance depends on how systems are designed for collaboration rather than accuracy alone.

Finally, recent work on table unionability showed that the task is difficult for both humans and LLMs [12], making it a particularly useful setting for studying assistance rather than automation. However, prior unionability work has primarily focused on benchmark construction, prediction quality, and comparative system performance, rather than on how users interact with AI support while making ambiguous decisions [8, 13, 14]. In parallel, the broader human–AI reliance literature has mostly studied generic classification or decision-support tasks rather than semantic data discovery tasks such as unionability [2, 11].

Our work fills this gap by bringing these strands together. We study unionability as an ambiguous human judgment task, and we examine how the *design* of LLM assistance — recommendation only, explanation, and explanation with uncertainty cues — shapes decision quality, confidence calibration, and users' willingness to revise their own initial judgments. In this sense, the contribution is not only to unionability research, but also to the HCI literature on appropriate reliance in domain-specific, ambiguity-rich decision settings.

3 Proposed System and Study Design

To investigate how LLM assistance should be introduced into an ambiguous semantic decision task, we design a survey-based prototype that supports both independent human judgment and subsequent AI-assisted revision. The study is intended not only to compare different forms of AI support, but also to evaluate a two-stage interaction design that allows human reasoning to be observed before and after exposure to LLM assistance.

3.1 System Design

We propose an LLM-assisted survey interface for table unionability assessment. The system is implemented as a survey containing nine table-pair questions, with three questions assigned to each AI-support condition: recommendation only, recommendation with explanation, and recommendation with explanation plus AI confidence. The survey is designed around a two-stage decision workflow. In the first stage, participants are shown a pair of tables and asked whether the tables are unionable. They provide an initial binary decision (Yes/No) and a confidence rating ranging from 0 to 100 using a slider. In the second stage, the survey reveals LLM-generated assistance, after which participants may revise their decision, update their confidence, and provide a short explanation. The pdf of the survey design is provided in our repository[1].

The nine table pairs were selected from a larger set of candidate unionability questions used in prior survey versions[12]. To construct the pilot, we first screened candidate items using two sources of difficulty information: prior human performance from the earlier survey and LLM responses collected in advance using a consistent prompt, see Example 3.1. Human difficulty was estimated based on question-level accuracy in the previous study [12]. LLM behavior was used to identify whether the model aligned with the benchmark label, how confident it was, and whether its explanation appeared strong, generic, or potentially misleading. Using these signals, we selected a balanced set of items spanning easier and harder cases, unionable and non-unionable cases, and both human-AI agreement and disagreement cases. This allows the pilot to test interface behavior across a range of realistic task conditions rather than only on trivially clear examples. You can find a detailed evaluation results in our repository. [2].

Example 3.1. Prompt Example

In the context of database management, a union combines two tables into a single table. Determining whether two tables are unionable may depend on semantics, structure, context, and interpretation, so some cases may be ambiguous. Evaluate the following two tables. Format your response exactly as:

- Unionable: Yes/No
- Explanation: 2–4 sentences
- Confidence: 0–100
- Difficulty: Easy/Medium/Ambiguous

Be concise and consistent across items.

The AI assistance panel varies by condition. In the recommendation-only condition, the survey presents the LLM's predicted answer. In the explanation condition, the survey presents both the LLM recommendation and a short natural-language rationale. In the confidence condition, the interface additionally presents the model's self-reported confidence percentage alongside the recommendation and explanation. The three conditions are interleaved across the nine questions so that participants experience multiple forms of AI support within a single survey. This design allows us to compare how different forms of LLM assistance shape human reasoning, confidence, and answer revision in ambiguous unionability tasks.

3.2 Method

We evaluate the proposed survey through a pilot user study designed to examine how different forms of LLM assistance influence unionability decisions. The study compares three interface conditions and analyzes whether the effects of AI support differ across easier and more ambiguous table-pair cases. Our goal is not only to assess whether AI assistance improves performance, but also to understand how interface design shapes reliance, confidence, and judgment revision in this task. The evaluation combines quantitative and qualitative measures.

Quantitative analysis focuses on four metrics. First, *decision accuracy* is measured as the proportion of correct unionability judgments before and after AI assistance, compared across conditions. Second, *confidence calibration* is assessed using the calibration measure from Shraga et al. [15]:

[1]https://github.com/NinaKlimenkova/LLM-Assistance-for-Table-Unionability
[2]https://github.com/NinaKlimenkova/LLM-Assistance-for-Table-Unionability

Table 1. Initial vs. Final Accuracy by Condition

Condition	n	Initial Accuracy	Final Accuracy	Δ
Recommendation Only	21	81.0%	81.0%	0.0 pp
Explanation	21	61.9%	76.2%	+14.3 pp
Uncertainty	21	61.9%	57.1%	-4.8 pp
Overall	**63**	**68.3%**	**71.4%**	**+3.1 pp**

$$\mathrm{Cal}(g) = \bar{c}_g - \mathrm{Acc}_g \tag{1}$$

where $\bar{c}_g$ is the mean reported confidence and Acc_g is the empirical accuracy for a group of decisions g. A calibration score closer to zero indicates better alignment between confidence and actual performance; positive values indicate overconfidence. We compute this measure per condition and compare pre-AI and post-AI calibration to evaluate whether different forms of assistance help users become more appropriately confident. Third, *decision transitions* are categorized into four outcomes—stable correct, corrected by AI, overcorrected, and persistent error—to characterize how AI assistance reshapes the distribution of correct and incorrect decisions rather than measuring accuracy change alone. Fourth, *mean confidence shift* is computed as the difference between post-AI and pre-AI confidence per condition, to assess whether certain forms of assistance inflate confidence without corresponding accuracy gains.

Qualitative analysis examines two sources. First, participants rate the *perceived helpfulness* of AI assistance on a 1–5 Likert scale after each question, providing a per-condition measure of subjective usefulness. Second, participants' *free-text explanations* are thematically coded to identify patterns in how users engage with AI support. Codes include *confirmed reasoning*, *no effect*, *changed mind*, *confidence boost*, *disagreed*, *critical of AI*, and *hesitation*. These themes are compared across conditions to examine whether richer forms of assistance encourage more reflective engagement versus passive acceptance or dismissal. All study procedures are designed with ethical research practices in mind, including informed consent and anonymized data collection.

3.3 Participants

We recruited 7 participants for the pilot study, all graduate students at Worcester Polytechnic Institute from fields including Computer Science, Data Science, Neuroscience, and Aerospace Engineering. We focused on technically literate users who are comfortable interpreting tabular data, as unionability assessment is most relevant to people who work with structured data, even if they are not database experts. Participants were recruited from classmates, lab members, and university networks. Each participant completed all 9 table-pair questions across the three AI assistance conditions, yielding 63 total observations. Self-reported English proficiency ranged from intermediate to fluent. The study was conducted in an online setting using a Qualtrics[3]-based survey prototype, with informed consent obtained prior to participation. The raw Qualtrics output is available in our repository for the community to use and build upon.[4].

4 Preliminary Results and Findings

This section presents the preliminary results and findings from the pilot study. We begin with interface recommendations gathered from participant feedback (Section 4.1), which highlight usability improvements for the next iteration of the

[3]https://wpi.qualtrics.com/jfe/form/SV_8jqmFQVl43NcHci
[4]https://github.com/NinaKlimenkova/LLM-Assistance-for-Table-Unionability

system. We then report a quantitative and qualitative performance analysis across the three AI assistance conditions (Section 4.2), examining decision accuracy, confidence calibration, answer revision patterns, and perceived helpfulness. Finally, we summarize the key takeaways from the pilot and their implications for the full study design (Section 4.3).

4.1 Interface Recommendations from users

Following the survey, participants provided feedback on the clarity and usability of the interface. Five recurring suggestions emerged. First, **keep table pairs visible** throughout the survey page, as the original tables disappeared when AI assistance was revealed, making it difficult to evaluate recommendations in context. Second, **remind users of their initial response** alongside the AI recommendation, since participants sometimes forgot their prior answer and confidence by Stage 2. Third, **allow backward navigation** so users can revisit earlier questions and reconsider responses. Fourth, **clarify what AI confidence represents**, as participants were unsure whether the percentage reflected certainty about unionability or about the recommendation's correctness. Fifth, **provide more context about table contents**, including additional sample rows or brief descriptions of each table's source and purpose. These recommendations highlight that the surrounding interface—not only the AI output itself—plays an important role in shaping user engagement with LLM assistance.

4.2 Performance Analysis

We analyzed pilot data across three dimensions: decision accuracy and transition patterns, confidence calibration, and qualitative engagement with AI assistance.

Decision accuracy. Overall accuracy improved modestly from 68.3% to 71.4% after AI assistance, but the effect varied considerably across conditions (Table 1). The "Explanation" condition showed the largest gain (+14.3pp, from 61.9% to 76.2%), while the "Uncertainty" condition slightly decreased accuracy (4.8pp, from 61.9% to 57.1%). "Recommendation Only" remained unchanged at 81.0%, though these questions had higher baseline difficulty. To better characterize how AI reshaped decisions, we categorized all 63 responses into four transition outcomes Figure 1: we see that 40 remained stable correct, 5 were corrected by AI, 3 were overcorrected (correct to incorrect), and 15 were persistent errors. Notably, the "Explanation" condition produced 3 corrections with zero overcorrections, while the "Uncertainty" condition overcorrected 2 responses and corrected only 1.

Confidence calibration. Using the calibration measure described in(Section 3.2), we examined whether AI assistance helped participants align their confidence with their actual accuracy. Figure 2 shows pre-AI and post-AI confidence for correct and incorrect answers by condition. The "Explanation" condition dramatically improved calibration: the gap between mean confidence on correct versus incorrect answers widened from 5.6pp to 34.5pp, indicating that participants became appropriately more confident when right and less confident when wrong. Recommendation Only also widened (31.3pp to 48.9pp), though these items were easier. Uncertainty worsened calibration (6.2pp to 3.7pp), suggesting that model confidence cues did not help participants distinguish correct from incorrect judgments. We also observed that "Recommendation Only" produced the largest raw confidence boost (+6.5) despite no accuracy change, raising a concern about unjustified confidence inflation.

Perceived helpfulness and qualitative themes. Participants rated the helpfulness of AI assistance on a 1–5 Likert scale after each question. Explanation ($\bar{x}$ = 3.00) and Uncertainty ($\bar{x}$ = 2.85) were rated substantially more helpful than Recommendation Only ($\bar{x}$ = 1.81). We also coded participants' 63 free-text explanations into seven themes (Figure 3). "No effect" dominated the Recommendation Only condition (12 of 26 codes), indicating that bare recommendations were largely dismissed. In contrast, "Confirmed reasoning" was the most frequent theme under Explanation (10 of 23 codes),

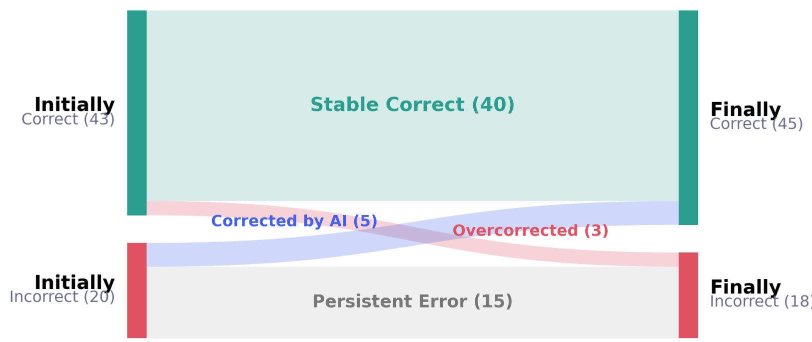

Fig. 1. Change in participant correctness before and after AI assistance.

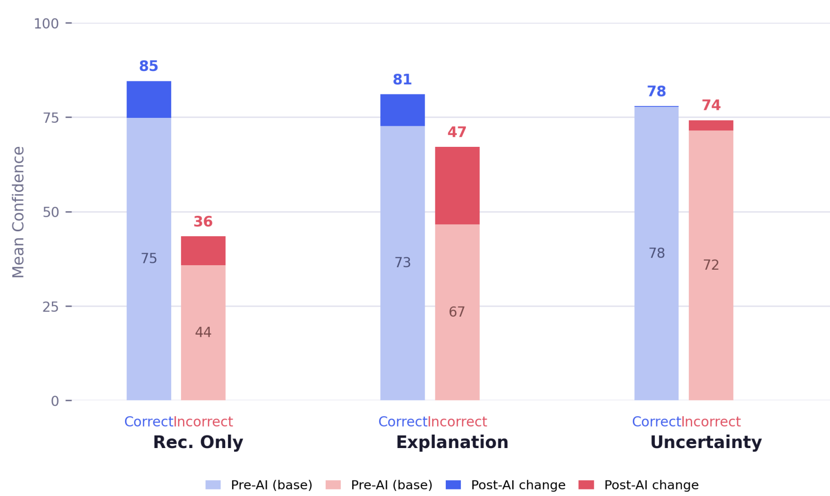

Fig. 2. Mean confidence before and after AI assistance by correctness and condition.

suggesting that explanations actively supported users in validating their own thinking. The Uncertainty condition uniquely produced "Hesitation" (2) and the highest count of "Changed mind" (4), consistent with the quantitative finding that uncertainty cues prompt active deliberation that does not always lead to better outcomes.

4.3 Key Takeaways

The pilot study yielded several preliminary insights that inform the research hypotheses and the design of the full study.

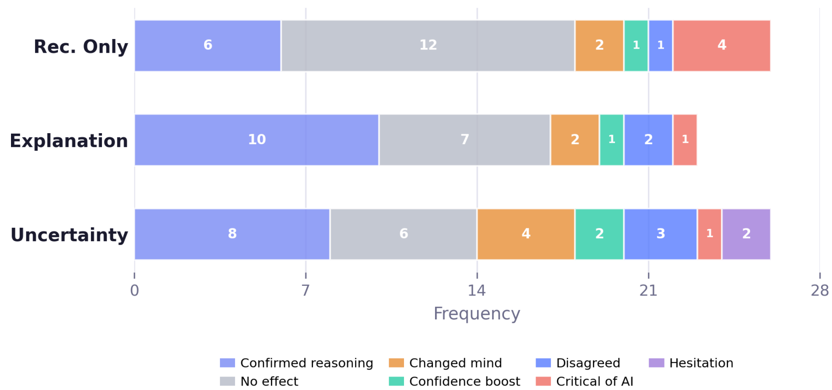

Fig. 3. Qualitative effects of AI assistance by condition.

The form of assistance matters more than its presence. AI assistance did not uniformly improve decisions. Explanations were the only condition that consistently helped—improving accuracy with no overcorrections. Uncertainty cues, despite providing the richest information, slightly worsened outcomes. This suggests that more information does not necessarily lead to better decisions.

Explanations support calibration; uncertainty cues do not. Explanations helped users become more confident when correct and less confident when incorrect. In contrast, uncertainty cues failed to improve calibration, and recommendation-only assistance inflated confidence without improving accuracy. This indicates that explanations help users assess the quality of their own reasoning, while model confidence scores do not achieve the same effect.

Users engage meaningfully only when given reasoning. Without justification, AI recommendations were largely dismissed. Explanations actively supported users in validating their own thinking, while uncertainty cues triggered deliberation that did not always lead to better outcomes. Users rated explanation-based assistance as most helpful.

5 Conclusion

This project positions table unionability as a human-AI interaction problem rather than only a technical matching task. By designing and piloting a two-stage LLM-assisted survey interface, we examined how different forms of AI assistance shape decision quality, confidence calibration, and revision behavior in ambiguous semantic data tasks.

Our pilot findings suggest that explanations are the most effective form of assistance: they improve both accuracy and calibration while encouraging selective revision. Recommendation-only assistance is largely ignored, and uncertainty cues can prompt deliberation that worsens outcomes. These results highlight that more AI transparency does not necessarily lead to better human decisions—what matters is how assistance is designed to support reasoning.

In future work, we plan to conduct a larger-scale study incorporating interface improvements and redesign our survey into a between-subject study. We also plan to investigate whether human-AI disagreement can serve as a signal of ambiguity in data integration tasks.

References

[1] Rakefet Ackerman, Avigdor Gal, Tomer Sagi, and Roee Shraga. 2019. A Cognitive Model of Human Bias in Matching. In *PRICAI 2019: Trends in Artificial Intelligence: 16th Pacific Rim International Conference on Artificial Intelligence, Cuvu, Yanuca Island, Fiji, August 26–30, 2019, Proceedings, Part I* (Cuvu, Yanuka Island, Fiji). Springer-Verlag, Berlin, Heidelberg, 632–646. doi:10.1007/978-3-030-29908-8_50

[2] Saleema Amershi, Dan Weld, Mihaela Vorvoreanu, Adam Fourney, Besmira Nushi, Penny Collisson, Jina Suh, Shamsi Iqbal, Paul N. Bennett, Kori Inkpen, Jaime Teevan, Ruth Kikin-Gil, and Eric Horvitz. 2019. Guidelines for Human-AI Interaction. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland Uk) *(CHI '19)*. Association for Computing Machinery, New York, NY, USA, 1–13. doi:10.1145/3290605.3300233

[3] Gagan Bansal, Besmira Nushi, Ece Kamar, Eric Horvitz, and Daniel S. Weld. 2021. Is the Most Accurate AI the Best Teammate? Optimizing AI for Teamwork. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 35. 11405–11414. doi:10.1609/aaai.v35i13.17359

[4] Alex Bogatu, Alvaro A. A. Fernandes, Norman W. Paton, and Nikolaos Konstantinou. 2020. Dataset Discovery in Data Lakes. In *2020 IEEE 36th International Conference on Data Engineering (ICDE)*. 709–720. doi:10.1109/ICDE48307.2020.00067

[5] Zana Buçinca, Maja Barbara Malaya, and Krzysztof Z. Gajos. 2021. To Trust or to Think: Cognitive Forcing Functions Can Reduce Overreliance on AI in AI-assisted Decision-making. *Proc. ACM Hum.-Comput. Interact.* 5, CSCW1, Article 188 (April 2021), 21 pages. doi:10.1145/3449287

[6] Shiye Cao, Anqi Liu, and Chien-Ming Huang. 2024. Designing for Appropriate Reliance: The Roles of AI Uncertainty Presentation, Initial User Decision, and User Demographics in AI-Assisted Decision-Making. *Proc. ACM Hum.-Comput. Interact.* 8, CSCW1, Article 41 (April 2024), 32 pages. doi:10.1145/3637318

[7] Valerie Chen, Q. Vera Liao, Jennifer Wortman Vaughan, and Gagan Bansal. 2023. Understanding the Role of Human Intuition on Reliance in Human-AI Decision-Making with Explanations. *Proc. ACM Hum.-Comput. Interact.* 7, CSCW2, Article 370 (Oct. 2023), 32 pages. doi:10.1145/3610219

[8] Yuhao Deng, Chengliang Chai, Lei Cao, Qin Yuan, Siyuan Chen, Yanrui Yu, Zhaoze Sun, Junyi Wang, Jiajun Li, Ziqi Cao, Kaisen Jin, Chi Zhang, Yuqing Jiang, Yuanfang Zhang, Yuping Wang, Ye Yuan, Guoren Wang, and Nan Tang. 2024. LakeBench: A Benchmark for Discovering Joinable and Unionable Tables in Data Lakes. *Proc. VLDB Endow.* 17, 8 (April 2024), 1925–1938. doi:10.14778/3659437.3659448

[9] Grace Fan, Jin Wang, Yuliang Li, Dan Zhang, and Renée J. Miller. 2023. Semantics-Aware Dataset Discovery from Data Lakes with Contextualized Column-Based Representation Learning. *Proc. VLDB Endow.* 16, 7 (March 2023), 1726–1739. doi:10.14778/3587136.3587146

[10] Aamod Khatiwada, Grace Fan, Roee Shraga, Zixuan Chen, Wolfgang Gatterbauer, Renée J. Miller, and Mirek Riedewald. 2023. SANTOS: Relationship-based Semantic Table Union Search. *Proc. ACM Manag. Data* 1, 1, Article 9 (May 2023), 25 pages. doi:10.1145/3588689

[11] Shuai Ma, Ying Lei, Xinru Wang, Chengbo Zheng, Chuhan Shi, Ming Yin, and Xiaojuan Ma. 2023. Who Should I Trust: AI or Myself? Leveraging Human and AI Correctness Likelihood to Promote Appropriate Trust in AI-Assisted Decision-Making. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) *(CHI '23)*. Association for Computing Machinery, New York, NY, USA, Article 759, 19 pages. doi:10.1145/3544548.3581058

[12] Sreeram Marimuthu, Nina Klimenkova, and Roee Shraga. 2025. Humans, Machine Learning, and Language Models in Union: A Cognitive Study on Table Unionability. In *Proceedings of the Workshop on Human-In-the-Loop Data Analytics* (Intercontinental Berlin, Berlin, Germany) *(HILDA '25)*. Association for Computing Machinery, New York, NY, USA, Article 6, 7 pages. doi:10.1145/3736733.3736740

[13] Koyena Pal, Aamod Khatiwada, Roee Shraga, and Renée J. Miller. 2024. ALT-GEN: Benchmarking Table Union Search using Large Language Models. In *VLDB Workshops*. https://vldb.org/workshops/2024/proceedings/TaDA/TaDA.3.pdf

[14] Ermu Qiu, Jun Gao, Yaofeng Tu, and Jingru Yang. 2025. LIFTus: An Adaptive Multi-Aspect Column Representation Learning for Table Union Search. *2025 IEEE 41st International Conference on Data Engineering (ICDE)* (2025), 2174–2187. https://api.semanticscholar.org/CorpusID:280693810

[15] Roee Shraga, Avigdor Gal, and Haggai Roitman. 2018. What Type of a Matcher Are You? Coordination of Human and Algorithmic Matchers. In *Proceedings of the Workshop on Human-In-the-Loop Data Analytics* (Houston, TX, USA) *(HILDA '18)*. Association for Computing Machinery, New York, NY, USA, Article 12, 7 pages. doi:10.1145/3209900.3209905

[16] Yunfeng Zhang, Q. Vera Liao, and Rachel K. E. Bellamy. 2020. Effect of confidence and explanation on accuracy and trust calibration in AI-assisted decision making. In *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency* (Barcelona, Spain) *(FAT* '20)*. Association for Computing Machinery, New York, NY, USA, 295–305. doi:10.1145/3351095.3372852

Received 20 February 2007; revised 12 March 2009; accepted 5 June 2009